BOUNDED ANTICANONICAL METRICS ON KLT PAIRS AND TORUS QUOTIENTS

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Abstract. Let (W, D) be a projective $\mathbb{Q}$-factorial klt pair with effective rational boundary, and put $P = -(K_W + D)$. We show that a positive Cartier multiple of P has a nonzero section provided that P is nef, that its pullback to a resolution carries a semipositive metric with locally bounded weights, and that the resolution has nonzero structure-sheaf Euler characteristic. Consequently a smooth rationally connected projective variety with smoothly semipositive anticanonical bundle has a nonzero anticanonical plurisection that is naturally invariant under every algebraic torus action.

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1. Introduction

The question. Fix a pair (W, D) and write $P = -(K_W + D)$ for the anti-log-canonical divisor. Nefness of P means that $P \cdot C \geq 0$ for every curve C. Nefness alone produces no divisor on W: it is a numerical shadow of positivity, not a section. The *nonvanishing problem* asks the sharper question: is $H^0(W, mP) \neq 0$ for some positive integer m? A "yes" means P is not only nef but carries genuine holomorphic functions. This paper answers "yes" under an explicit analytic hypothesis, and then applies the answer to quotients by torus actions.

The criterion. All varieties are integral and defined over $\mathbb{C}$.

Theorem 1.1 (Bounded-metric nonvanishing)**.** *Let (W, D) be a projective $\mathbb{Q}$-factorial klt pair with effective rational boundary, and let $P = -(K_W + D)$. Assume:*

(i) *P is nef;*
(ii) *on a smooth projective resolution $\rho : \widetilde{W} \to W$, the pullback $\rho^* P$ carries a semipositive metric whose weights are locally bounded;*
(iii) *$\chi(\widetilde{W}, \mathcal{O}_{\widetilde{W}}) \neq 0$.*

Then $H^0(W, \mathcal{O}_W(mP)) \neq 0$ for some integer $m > 0$ with mP Cartier.

The hypotheses play distinct roles. Nefness lets us pair with curves and push classes; the bounded semipositive metric is exactly the positivity that hard Lefschetz and adjoint direct-image theory consume; the Euler characteristic is the spark that produces twisted differential forms. The $\mathbb{Q}$-factoriality and the effective boundary enter the birational numerical arguments and the cotangent-subsheaf step respectively; both are used and neither is hidden. All three hypotheses are stable under passing to a common higher resolution, because bounded psh weights pull back and $\chi(\mathcal{O})$ is a birational invariant of smooth projective varieties.

The application. The point of controlling the boundary is that the theorem survives quotienting by a torus.

Theorem 1.2 (Invariant sections)**.** *Let Z be a smooth rationally connected projective variety whose anticanonical bundle has a smooth semipositive Hermitian metric. Then for every algebraic torus T acting on Z there is an integer $m > 0$ with*

$$H^0(Z, -mK_Z)^T \neq 0,$$

the invariants taken with respect to the natural *anticanonical linearization induced by the differential of the action on T_Z.*

The word "natural" is essential. To build the quotient we perturb the anticanonical class by a small ample class and a small character; this auxiliary linearization only selects an open set. The linearization in the conclusion is the differential one and is never shifted. A character twist changes the answer; Remark 3.6 records the elementary example on $\mathbb{P}^1$.

Together with the finite-cover structure theorem of the companion article [OpenAI(2026)], Theorem 1.2 yields Corollary 4.1: for every smooth connected projective variety X with smoothly semipositive anticanonical bundle, $H^0(X, -mK_X) \neq 0$ for some $m > 0$. The companion proves Corollary 4.1 directly by a finite-volume argument; here we route it through the bounded-metric criterion and a stable quotient, borrowing only the companion's finite-cover structure and finite-étale norm lemma. The positive multiple cannot be reduced to the first power in general: an Enriques surface has $H^0(S, -K_S) = 0$, and taking the product with $\mathbb{P}^1$ gives a smoothly semipositive example with no first-power section (§4).

How the proofs run. *Theorem 1.1.* Eight steps.

(1) **Twisted forms.** Riemann–Roch and the nonzero Euler characteristic produce, for infinitely many twists $t \to \infty$, a nonzero section of $\Omega^p_{\widetilde{W}} \otimes \mathcal{O}(t\rho^* P)$ (hard Lefschetz with multiplier ideals).
(2) **Determinant.** Determinants of these forms give effective divisors $N_i \sim_{\mathbb{Q}} M + m_i P$, $m_i \to \infty$, with $-M$ pseudoeffective.
(3) **Maximal fibration.** Choose a fibration $\pi : S \to W$, $f : S \to Y$ of maximal base dimension whose polarization is pseudoeffectively dominated by $\pi^* P$; maximality forbids any extra pencil controlling the general fiber.
(4) **Decomposition.** The N_i split as $r\pi^* P = f^* V + H$ with V a line bundle on Y and H effective, so that over every prime of Y some component of $f^* G$ is missing from H.
(5) **Rounding.** Discrepancy rounding of the klt pair gives $E = \lceil R \rceil$, $\Xi = \lceil R \rceil - R$, and $K_S + \pi^* P + \Xi = E$; maximality gives rank-one adjoint direct images.
(6) **Fiber-integral metric.** Normalized L^a-integrals of a section of H converge to a fiberwise essential supremum, giving a bounded semipositive metric on V; the missing components bound it at the boundary.
(7) **Vanishing.** A small mixture with a strictly positive metric (Skoda) keeps the multiplier ideal trivial, so Fujino's Kollár–Nadel vanishing applies to every nonnegative twist.
(8) **Euler polynomial.** Vanishing for all twists makes $\chi(Y, \mathcal{G} \otimes aV)$ a polynomial positive at $a = 0$; some positive value is nonzero, whence a section of mP on W.

Theorem 1.2. An ample perturbation with a small character makes the stable locus equal the semistable one, and $W = U/T$ is projective. Luna slices exhibit a klt boundary $D = \sum(1 - 1/e_B)B$ and identify $-K_Z|_U = q^* P$ with the natural linearization. Convexity along algebraic arcs shows translated lifts have no poles; compactifying the action graph upgrades this to a locally uniform bound, which makes the orbit-supremum metric bounded and psh (Kiselman). Rational connectedness gives $\chi = 1$ on a resolution of W. Theorem 1.1 then yields a section, the bounded quotient metric bounds its pullback, and removable singularities extend it over the complement of U as an invariant global section.

Context. Two notions must be separated. A divisor is *num-effective* if it is numerically equivalent to an effective real divisor; a *section of a positive multiple* asks for linear equivalence. The two differ exactly where our analytic hypothesis enters. Lazić–Matsumura–Peternell–Tsakanikas–Xie [Lazić–Matsumura–Peternell–Tsakanikas–Xie(2023)] prove num-effectivity of nef anti-log-canonical divisors on projective log-canonical threefolds that are $\mathbb{Q}$-factorial or have rational singularities; their Theorem E, under a nonzero $\chi(\mathcal{O})$ hypothesis, is the closest predecessor to Theorem 1.1. We upgrade num-effectivity to an actual section in arbitrary dimension, at the price of the bounded metric. Müller [Müller(2025)] proves ordinary nonvanishing for klt pairs with semiample anticanonical divisor on general rationally connected fibers (all dimensions) and unconditionally in dimension three, plus an equivariant statement for commutative groups. Our equivariant theorem reaches the same invariant conclusion with a different positivity hypothesis (smooth semipositivity rather than semiampleness), restricted to smooth rationally connected varieties and torus actions. That smooth semipositivity is genuinely weaker already appears in the surface classification of Chen–Filip–Sun–Tosatti–Zhang [Chen–Filip–Sun–Tosatti–Zhang(2026)]: rational surfaces with smoothly semipositive anticanonical bundle and a smooth elliptic anticanonical curve of nontorsion degree-zero normal bundle, whose anticanonical bundle is not semiample.

The technical ancestry: hard Lefschetz with multiplier ideals is the mechanism of Demailly–Peternell–Schneider [Demailly–Peternell–Schneider(2001)]; the determinant step is Lazić–Peternell [Lazić–Peternell(2018)], adapted to anticanonical geometry in [Lazić–Matsumura–Peternell–Tsakanikas–Xie(2023)]

whose Theorem 4.1 controls the negative first Chern class of the saturated cotangent-tensor subsheaf. Positivity transfers to the base of a fibration through adjoint direct images of Berndtsson [Berndtsson(2009)], Berndtsson–Păun [Berndtsson–Păun(2008)], and Păun–Takayama [Păun–Takayama(2018)]; we use the singular version because rounding leaves fractional divisorial poles even when the metric is bounded. Skoda's integrability theorem [Skoda(1972)] and Fujino's Kollár–Nadel vanishing [Fujino(2018)] let one metric construction govern all twists at once. On the quotient side we use projective GIT [Mumford–Fogarty–Kirwan(1994)], torus linearization [Sumihiro(1974), Knop–Kraft–Luna–Vust(1989)], attracting cells of Białynicki-Birula [Białynicki-Birula(1973)], Luna's slice theorem [Luna(1973)], and Kiselman's minimum principle [Kiselman(1978), Deng–Wang–Zhang–Zhou(2019)]. The smooth application draws on Yau's prescribed-Ricci theorem [Yau(1978)], the structure theorem of Demailly–Peternell–Schneider [Demailly–Peternell–Schneider(1996)], and Campana–Demailly–Peternell [Campana–Demailly–Peternell(2015)].

Plan. Section 2 proves Theorem 1.1. Section 3 builds the stable quotient and proves Theorem 1.2. Section 4 proves Corollary 4.1 and the Enriques example.

2. A bounded-metric criterion for klt pairs

Throughout, (W, D) satisfies the hypotheses of Theorem 1.1 and $P = -(K_W + D)$. Equalities of line bundles mean linear equivalence; equalities of rational divisors mean $\mathbb{Q}$-linear equivalence. A statement about an effective divisor, its support, or a coefficient is a statement about the actual divisor; we keep these uses apart.

2.1. Analytic and numerical tools.

We record the imported results in the exact forms used.

Hard Lefschetz with multiplier ideals. Let B be a line bundle on a compact Kähler n-fold S with a semipositive singular metric h of weight φ. Then [Demailly–Peternell–Schneider(2001), Theorem 2.1.1] gives a surjection

$$H^0(S, \Omega_S^{n-q} \otimes B \otimes \mathcal{I}(h)) \longrightarrow H^q(S, K_S \otimes B \otimes \mathcal{I}(h)), \tag{1}$$

where $\mathcal{I}(h)$ tests local integrability of $|g|^2 e^{-\varphi}$ in a frame. In particular every positive power of a *bounded* metric has trivial multiplier ideal.

Singular adjoint direct images. For a projective surjective morphism $f : S \to Y$ of smooth varieties with connected fibers and a semipositive singular metric on B, the theorem of Păun–Takayama [Păun–Takayama(2018), §3.2.1 and Theorems 3.3.4–3.3.5] applies: if the multiplier ideal is trivial on the total space and on general fibers, the fiberwise L^2 metric on $f_*(K_{S/Y} + B)$ has semipositive curvature in the singular sense. On a base-change open where the direct image has rank one and its generator has finite integral, the negative logarithm of that integral is a psh weight. The integral agrees with the metric almost everywhere; no continuity or global base change is needed. We apply this for each integer twist separately, so the base-change open may depend on the twist.

Kollár–Nadel vanishing. The theorem of Fujino [Fujino(2018), Theorem 1.3]: if $f : S \to Y$ is a surjective morphism from a compact Kähler manifold to a projective variety and a singular metric on B has curvature at least $\epsilon f^*\omega_A$ for some $\epsilon > 0$ and a positive curvature form of an ample line bundle A on Y, then

$$H^i(Y, R^j f_*(K_S \otimes B \otimes \mathcal{I}(h))) = 0 \qquad (i > 0,\ j \geq 0). \tag{2}$$

Only $j = 0$ is used.

A numerical fact. Because W is $\mathbb{Q}$-factorial, pushforward along a birational morphism $\pi : S \to W$ (with S smooth) preserves pseudoeffectivity; compare [Lazić–Matsumura–Peternell–Tsakanikas–Xie(2023), Lemma 3.4]. Indeed, if F is numerically trivial on S, then $\pi^*\pi_*F - F$ is exceptional and numerically trivial over W; the negativity lemma applied to it and its negative forces it to vanish. Intersecting $\pi^*\pi_*F = F$ with curves dominating curves of W makes π_*F numerically trivial. Hence π_*

is a well-defined continuous linear map on finite-dimensional numerical class groups, sending effective classes to effective classes and preserving their closed cone. Pullback of a pseudoeffective $\mathbb{Q}$-Cartier class by the birational maps used below also stays pseudoeffective.

2.2. **Twisted forms and the determinant.** The first goal is an unbounded sequence of effective divisors differing from one fixed class by multiples of P. We adapt the determinant argument of Lazić–Peternell [Lazić–Peternell(2018), Lemma 4.1], as used in [Lazić–Matsumura–Peternell–Tsakanikas–Xie(2023 Lemma 5.1]. Here the twisting exponents are already positive and unbounded, so a basis containing a high-exponent section has an unbounded sum of exponents; this lets us avoid importing the numerical-nontriviality hypothesis of the cited lemma.

Lemma 2.1. *Either $H^0(W, mP) \neq 0$ for some positive Cartier multiple, or there are a Weil divisor M, strictly increasing positive integers m_i, and effective Weil divisors N_i with*

$$N_i \sim_{\mathbb{Q}} M + m_i P, \qquad -M \in \overline{\mathrm{Eff}}(W), \tag{3}$$

all $m_i P$ Cartier.

Proof. The case W a point is immediate, so take $\dim W = n > 0$. Choose a smooth projective resolution $\rho : \widetilde{W} \to W$ on which the metric is bounded, and $r_0 > 0$ with $r_0 P$ Cartier.

Step 1 (Riemann–Roch). The function $t \mapsto \chi(\widetilde{W}, K_{\widetilde{W}} + t\rho^*(r_0 P))$ is a polynomial in t. At $t = 0$ it equals $(-1)^n \chi(\widetilde{W}, \mathcal{O}_{\widetilde{W}}) \neq 0$ by hypothesis, so it is nonzero for all but finitely many positive integers t. For such t at least one cohomology group of $K_{\widetilde{W}} + t\rho^*(r_0 P)$ is nonzero.

Step 2 (hard Lefschetz). Apply (1) with $B = t\rho^*(r_0 P)$ and its bounded semipositive metric, whose multiplier ideal is trivial. If $H^q \neq 0$, we get a nonzero section

$$\sigma \in H^0\big(\widetilde{W}, \Omega^{n-q}_{\widetilde{W}} \otimes \mathcal{O}(t\rho^* P)\big).$$

There are finitely many cohomological degrees q, so a single fixed q, and hence a fixed $p = n - q$, occurs for an infinite sequence $t = n_i \to \infty$, giving nonzero sections

$$\sigma_i \in H^0\big(\widetilde{W}, \Omega^p_{\widetilde{W}} \otimes \mathcal{O}(n_i \rho^* P)\big), \qquad n_i \to \infty.$$

Step 3 (the $p = 0$ case). If $p = 0$, projecting to W and using $\rho_* \mathcal{O}_{\widetilde{W}} = \mathcal{O}_W$ gives $H^0(W, \mathcal{O}(n_i r_0 P)) \neq 0$, the first alternative.

Step 4 (the determinant line). Now $p > 0$. Work over the function field $K = \mathbb{C}(W)$. Trivialize $r_0 P$ rationally, so untwisting the σ_i produces vectors in $\bigwedge^p \Omega^1_{K/\mathbb{C}}$. Let F_K be their K-span, of dimension $b > 0$. Its determinant is a line

$$\det F_K \;\subset\; \bigwedge^b \Big(\bigwedge^p \Omega^1_{K/\mathbb{C}} \Big) \;\subset\; (\Omega^1_{K/\mathbb{C}})^{\otimes pb},$$

the last inclusion via alternating tensors in characteristic zero. Intersect this line with the reflexive tensor power $(\Omega^1_W)^{[\otimes pb]}$ and take the saturated reflexive rank-one extension; it is $\mathcal{O}_W(M)$ for a Weil divisor M.

Step 5 (a section). The determinant of any b of the untwisted vectors is a nonzero section of $\mathcal{O}_W(M + \sum n_i P)$ on the smooth locus of W, giving the identity. It is regular at every codimension-one point, since ρ is an isomorphism there and the σ_i are holomorphic; reflexivity extends it across codimension two. Its zero divisor N is an effective Weil divisor, so $N \sim_{\mathbb{Q}} M + \sum n_i P$ for the chosen subset of exponents.

Step 6 (unbounded exponents). The exponent sums are unbounded: fix any nonzero vector at an arbitrarily high twist n_i and extend it to a basis; all exponents are positive, so the sum is at least n_i. Selecting a strictly increasing subsequence of these sums gives the m_i, effective divisors N_i, and (3). Finally, (W, D) is projective klt with effective boundary and $-(K_W + D) = P$ nef, so

[Lazić–Matsumura–Peternell–Tsakanikas–Xie(2023), Theorem 4.1] applies to the saturated rank-one subsheaf $\mathcal{O}_W(M)$ of the reflexive cotangent tensor power: its negative first Chern class is pseudoeffective. Since W is $\mathbb{Q}$-factorial, $-M$ is pseudoeffective, giving $-M \in \overline{\mathrm{Eff}}(W)$. □

2.3. **A maximal dominated fibration.** Assume now the second alternative of Lemma 2.1. The next goal is to write a multiple of a birational pullback of P as a pullback from a base plus an effective divisor, with a component missing over each base prime. That missing component will keep the induced base metric free of a divisorial pole. To get it, take a base of maximal dimension whose polarization is pseudoeffectively dominated by P; maximality will forbid pencils that are nonconstant on a general fiber.

Consider diagrams

$$\begin{array}{ccc} S & \xrightarrow{\ f\ } & Y \\ {\scriptstyle\pi}\downarrow & & \\ W & & \end{array} \tag{4}$$

with S, Y smooth projective, π birational, f surjective, and

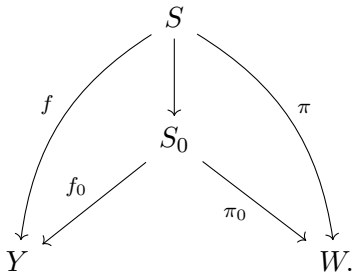
$$d\,\pi^*P - f^*A \in \overline{\mathrm{Eff}}(S) \tag{5}$$

for some integer $d > 0$ and ample line bundle A on Y; the map to a point is allowed, with the polarization omitted. Choose a diagram with $\dim Y$ maximal.

We may require connected general fibers. Passing to the relative algebraic closure of $\mathbb{C}(Y)$ in $\mathbb{C}(W)$ is a generically finite change of base; the pullback of A there is big and, on a smooth projective model, dominates a small ample class, so (5) persists after scaling, and smooth birational changes of the base are harmless. With $\mathbb{C}(Y)$ algebraically closed in $\mathbb{C}(W)$, the finite map in the Stein factorization of f is birational, hence an isomorphism as Y is normal. Thus f has connected fibers and $f_*\mathcal{O}_S = \mathcal{O}_Y$, as the direct-image theorem needs.

We also use a normal equidimensional intermediate model

(6)



Here π_0 is birational and S resolves S_0 and the pair on W. Construction: start from a projective graph of the rational map, flatten the main component by a projective modification of the base [Raynaud–Gruson(1971)], replace the base by a smooth projective modification, and normalize the main flat transform; finite normalization preserves equidimensionality over the smooth base. Finally resolve above this model, also dominating the resolution on which the metric is bounded. The smooth space S need not be equidimensional over Y; equidimensionality is used only on S_0, where it forces every vertical prime to dominate a prime of Y.

Lemma 2.2 (Excluding a new pencil)**.** *In a maximal diagram (4), let B be a line bundle on a birational model over S. If its class is pseudoeffectively dominated by $c\,\pi^*P + E'$ for some $c > 0$ and a π-exceptional divisor E', then no pencil in $|B|$ is nonconstant on the general fiber of f. The conclusion persists after adding to B the pullback of any fixed divisor on Y.*

Proof. Resolve the pencil, remove its fixed part, and combine its map to $\mathbb{P}^1$ with f. If the pencil is nonconstant on the general fiber, the combined image has dimension $> \dim Y$. A product polarization gives a basepoint-free line bundle F on the resolved source, pseudoeffectively dominated by a multiple of π^*P plus an exceptional divisor, using (5).

Choose a general member F_0 of that system; it has no π-exceptional component. Push the class comparison to W using the numerical fact of §2.1; the exceptional term vanishes and $c'P - \pi_* F_0$ is pseudoeffective. Pulling back, $c'\pi^*P - \pi^*\pi_* F_0$ is pseudoeffective. As $\pi_* F_0$ is effective $\mathbb{Q}$-Cartier, its pullback contains the strict transform F_0 plus an effective exceptional remainder, so $c'\pi^*P - F_0$ is pseudoeffective. This dominates a polarization of the larger projective image; on a smooth model the pulled-back polarization is big, hence dominates a small ample class, and after scaling and pulling back gives a diagram of the allowed form. That contradicts maximality.

For the final assertion, choose b with bA minus the fixed base divisor ample; the extra pullback is then dominated by bf^*A, hence by $bd\,\pi^*P$ via (5). Increase c. □

2.4. **Decomposition with a missing component.** Pull the N_i to S_0 (effective $\mathbb{Q}$-Cartier). The next lemma produces the decomposition controlling the base metric.

Lemma 2.3. *For some $r > 0$ with rP Cartier there are a line bundle V on Y and an effective Cartier divisor H on S with*

$$r\,\pi^*P = f^*V + H, \tag{7}$$

and for each prime G of Y some component of f^*G dominating G has coefficient zero in H.

Proof. For $m_i < m_j < m_k$ put

$$\alpha = \frac{m_k - m_j}{m_k - m_i}, \qquad \beta = \frac{m_j - m_i}{m_k - m_i},$$

so $\alpha, \beta > 0$, $\alpha + \beta = 1$, and $N_j \sim_{\mathbb{Q}} \alpha N_i + \beta N_k$. Clearing denominators, the two effective divisors give sections of one line bundle; that bundle is pseudoeffectively dominated by a multiple of P because $-M$ is pseudoeffective in (3). By Lemma 2.2 their ratio is constant on the general fiber. The ratio lies in $\mathbb{C}(Y)$ (relative algebraic closure), so its principal divisor has no horizontal component on S_0.

Hence the horizontal parts of $\pi_0^* N_i$ depend affinely on m_i as actual divisors. Fixing the first two indices writes them as $A_0 + m_i B_0$ with $\operatorname{Supp} A_0, \operatorname{Supp} B_0$ in one finite union of primes. Effectivity for unbounded m_i forces $B_0 \geq 0$. So the difference of two members divided by the difference of exponents is $\mathbb{Q}$-linearly equivalent to $\pi_0^* P$ with effective horizontal part. Clearing denominators gives a rational section of a positive Cartier multiple of $\pi_0^* P$ with no horizontal poles; write its divisor as T_0.

For each base prime G set

$$v_G = \min_{F \mapsto G} \frac{\operatorname{coeff}_F(T_0)}{\operatorname{coeff}_F(f_0^* G)},$$

a minimum over the finitely many primes F above G (denominators positive); only finitely many v_G are nonzero. Every vertical prime of S_0 sits above a base prime by equidimensionality, so

$$T_0 - f_0^*\Big(\sum_G v_G\, G\Big) \geq 0.$$

The difference is $\mathbb{Q}$-Cartier (T_0 is Cartier up to the chosen multiple, Y smooth). Scaling makes the base divisor integral and the remainder Cartier, giving V and an effective Cartier remainder on S_0; pull to S to get (7). The component attaining the minimum has remainder coefficient zero, and its strict transform on S still dominates G with coefficient zero in H. □

If Y is a point, (7) immediately gives an effective divisor linearly equivalent to $r\pi^*P$, hence a section downstairs. Henceforth $\dim Y > 0$. We build a semipositive metric on V from fiber integrals, and use the missing components to bound it. The integrals require adjoint line bundles whose fiberwise section spaces have dimension one; discrepancy rounding supplies these, and maximality supplies the rank-one property.

2.5. **Rounding and rank one for every twist.** With compatible canonical divisors put

$$R = K_S - \pi^*(K_W + D), \qquad E = \lceil R \rceil, \qquad \Xi = \lceil R \rceil - R. \tag{8}$$

Choose the resolution so $\operatorname{Supp} R$ is simple normal crossings. As the pair is klt, every coefficient of R exceeds -1. Along a nonexceptional prime the coefficient of R is minus its coefficient in D, which lies in $(-1, 0]$, so its ceiling is zero; thus E is integral, effective and π-exceptional. The divisor Ξ is effective with simple normal crossings support and coefficients in $[0, 1)$. The key identity is

$$K_S + \pi^*P + \Xi = E. \tag{9}$$

Lemma 2.4. *For every integer $a \geq 0$,* $\operatorname{rank} f_*\mathcal{O}_S(E + ar\pi^*P) = 1$.

Proof. Over the generic fiber the section of $E+aH$ (from (7)) is nonzero, including $a = 0$. Were the generic-fiber section space at least two-dimensional, twisting by a sufficiently ample pullback from Y would extend two independent generic sections to global sections whose ratio is nonconstant on the general fiber. Their line bundle is dominated by a multiple of π^*P plus the exceptional E, together with the base twist as in Lemma 2.2, contradicting that lemma. □

For $a \geq 0$ define the integral line bundle

$$B_a = (ar + 1)\pi^*P + \Xi = E + ar\pi^*P - K_S. \tag{10}$$

The second form proves integrality although each summand of the first can be rational. Equip B_a initially with weights $(ar + 1)\varphi_P + \varphi_\Xi$, where φ_P are the bounded weights on π^*P and φ_Ξ is the divisorial weight. Its curvature is semipositive and its multiplier ideal is trivial, because the weights of P are bounded and the coefficients of the snc divisor Ξ are below one. The same holds on general fibers after shrinking to where the boundary is relatively normal crossings.

2.6. **A bounded metric from normalized fiber integrals.**

Proposition 2.5. *V carries a singular Hermitian metric whose local weights are psh and locally bounded on Y.*

Proof. Fix a dense Zariski open Y° over which f is smooth with connected fibers, Ξ is relatively snc, and no fiber lies in $E \cup H$. In a coordinate neighborhood in Y° choose frames v of V and η of $-K_Y$. The section $s = f^*v\, s_H$ of $r\pi^*P$ has divisor H there, and $s_E s^a f^*\eta$ belongs to $K_{S/Y} + B_a$. By Lemma 2.4 it generates the direct image at a general point; on its base-change open this generator identifies the adjoint direct image with $aV - K_Y$. Since the direct image has rank one, its semipositive L^2 metric has psh weight equal to minus the logarithm of the squared L^2 norm of this generator. Dividing by a gives

$$u_a(y) = -\frac{1}{a}\log \int_{S_y} t^a \, d\mu_y, \qquad t = |s|^2_{h^r_P}, \quad a \geq 1. \tag{11}$$

The measure μ_y is the squared adjoint norm density of $s_E f^*\eta$ for the metric on $\pi^*P + \Xi$; locally it is a bounded positive factor (a nonzero holomorphic section squared) times $\prod_j |z_j|^{-2c_j}$ with $c_j < 1$. It is finite on a general compact fiber and has the same null sets as smooth volume; zeros of s_E and divisorial poles are measure zero. Only boundedness of φ_P is used.

Upper bound on Y°. For $y_0 \in Y^\circ$ pick a point of S_{y_0} off $E \cup H \cup \operatorname{Supp}\Xi$. On a small product neighborhood for the smooth map the holomorphic factors are bounded away from zero and the

metric weights are bounded, giving a neighborhood of y_0, constants $c, b > 0$, and open pieces in its fibers with $t \geq c$ and μ_y-mass $\geq b$. Thus $\int_{S_y} t^a \, d\mu_y \geq bc^a$ and $u_a(y) \leq -\log c - \frac{\log b}{a}$. These estimates continue the u_a across each exponent-dependent base-change exceptional set and bound them locally uniformly above.

The limit. Let Y^* be the countable intersection of the full-measure sets where the integral formula agrees with the psh representatives for every a, a common full-measure subset of Y° (not a common Zariski open). For $y \in Y^*$, the elementary L^a-norm limit on (S_y, μ_y) gives

$$\lim_{a\to\infty} u_a(y) = -\log \operatorname{ess\,sup}_{S_y} |s|^2_{h_P^r} =: u(y). \tag{12}$$

Indeed $(\int t^a)^{1/a} \leq M_y \, \mu_y(S_y)^{1/a}$ with $M_y = \operatorname{ess\,sup} t$, and for any $c < M_y$ the set $\{t > c\}$ has positive measure, giving $\liminf (\int t^a)^{1/a} \geq c$; let $c \uparrow M_y$.

A psh representative. The u_a are locally uniformly bounded above on Y° (properness, holomorphicity of s, bounded φ_P). They are locally bounded below: $\mu_y(S_y)$ is bounded above, since snc charts give density $\leq C \prod_j |z_j|^{-2c_j}$ with $c_j < 1$ and properness gives a finite cover above a compact base neighborhood. By local compactness of psh functions a subsequence converges in L^1_{loc}, and (12) identifies every subsequential limit a.e., so u has a psh representative. Under $v' = gv$, t scales by $|g|^2$ and u by $-\log|g|^2$; changing η contributes a term divided by a, vanishing in the limit. Thus u defines a metric on V over Y°.

Lower bound at the boundary. Positivity is not enough; we need local boundedness on all of Y. On the inverse image of a relatively compact base neighborhood, properness and bounded h_P bound $|s|^2$ above, so (12) bounds u below even near a missing base fiber. For an omitted prime $G \subset Y$, choose the component F above G on which H has coefficient zero. At a general point of F, $s \neq 0$, $F \to G$ is submersive, and transverse coordinates read $z_1 \circ f = w_1^e \, a(w)$ with $a(w) \neq 0$. Together with coordinates along G, nearby general fibers meet a neighborhood of that point in a nontrivial open set. On a smaller neighborhood $|s|^2_{h_P^r} \geq c > 0$ (the section is nonzero, h_P bounded), and those openings have positive μ_y-measure (same null sets as volume). A uniform lower bound on the measures is not needed for an essential supremum. Thus the component may lie in E or Ξ without harm, because on nearby general fibers those supports are measure zero; hence $\operatorname{ess\,sup} \geq c$ and $u \leq -\log c$ near a dense open subset of G.

Extension. The psh removable-singularity theorem extends u across these codimension-one sets using the above upper bound; the remainder is an analytic set of codimension at least two in the smooth base, and psh Hartogs extends u there too. The lower bound survives by upper semicontinuity and the extension has a local upper bound. The local extensions are unique and respect frame changes, giving a bounded metric on V over all of Y. This is local boundedness; no continuity of the original metric or of its essential supremum is asserted. □

2.7. **Mixing in small strict positivity.** The bounded metric on V lets us write every nonnegative integral twist $ar\pi^*P = af^*V + aH$ with a semipositive metric whose only unbounded weights come from the integral divisor aH, giving exactly the ideal $\mathcal{O}_S(-aH)$ that cancels aH. Vanishing still needs strict positivity in base directions, which we place on the fixed summand $\pi^*P + \Xi$ so it also serves the zeroth twist.

Lemma 2.6. *There is a metric on $\pi^*P + \Xi$ with curvature at least $\epsilon f^*\omega_A$ for some $\epsilon > 0$ and with trivial multiplier ideal.*

Proof. By (5), $d\pi^*P - f^*A$ carries a positive-current metric. Tensoring with a positive metric on f^*A and taking a dth root gives a metric on π^*P with weights ψ and curvature at least $d^{-1}f^*\omega_A$. For $0 < \tau < 1$ mix with the bounded metric:

$$\varphi_\tau = (1-\tau)\varphi_P + \tau\psi,$$

whose curvature is at least $(\tau/d)f^*\omega_A$. We find one small τ making $\varphi_\tau+\varphi_\Xi$ have trivial multiplier ideal. Choose $p>1$ with $pc_j<1$ for every coefficient c_j of Ξ and let $q=p/(p-1)$. On a log-resolution chart $e^{-p\varphi_\Xi}\asymp\prod_j|z_j|^{-2pc_j}$ is integrable; Skoda's small-exponent theorem [Skoda(1972)] gives $e^{-q\tau\psi}$ integrable for small τ. A finite cover lets one τ serve all charts. Hölder makes $e^{-(\varphi_\Xi+\tau\psi)}$ integrable, and the bounded factor $e^{-(1-\tau)\varphi_P}$ does not change integrability. This is exactly triviality of the multiplier ideal; no analytic-singularity assumption on ψ is used. □

2.8. **Proof of Theorem 1.1.**

Proof. Earlier arguments dispatch $\dim W=0$, the $p=0$ case of Lemma 2.1, and a zero-dimensional maximal base. So take V,H,E,Ξ as constructed, and h_V the bounded semipositive metric on V from Proposition 2.5. For each $a\ge0$, equip B_a with the tensor product of the metric of Lemma 2.6, the pullback $h_V^{\otimes a}$, and the divisor metric of aH. With g_H a local equation of H, its weight is

$$\varphi_{\mathrm{mix}}+af^*\varphi_V+a\log|g_H|^2,$$

of curvature at least the same positive multiple of $f^*\omega_A$. Because φ_V is bounded and aH integral, the multiplier ideal is exactly $\mathcal{O}_S(-aH)$: at a generic point of a component of H the remaining weight is bounded above, so integrability forces divisibility by g_H^a, and normality extends the quotient across codimension two; conversely after dividing the test is the trivial-ideal test of Lemma 2.6, unaffected by bounded weights.

The adjoint identity becomes $K_S+B_a-aH=E+af^*V$. Apply (2) with the projection formula. Writing $\mathcal{G}=f_*\mathcal{O}_S(E)$, for every $i>0$ and $a\ge0$,

$$H^i(Y,\mathcal{G}\otimes\mathcal{O}_Y(aV))=0. \tag{13}$$

The sheaf $\mathcal{G}$ is coherent of rank one (Lemma 2.4). Riemann–Roch on the smooth projective base makes $Q(a)=\chi(Y,\mathcal{G}\otimes\mathcal{O}_Y(aV))$ a polynomial in a. The effective E gives a nonzero section of $\mathcal{G}$, so (13) at $a=0$ gives $Q(0)=h^0(Y,\mathcal{G})>0$; the polynomial is nonzero, so some positive integer a has $Q(a)\neq0$. Again by (13), $Q(a)=h^0(Y,\mathcal{G}\otimes\mathcal{O}_Y(aV))\neq0$, hence

$$H^0(S,E+af^*V)\neq0.$$

Multiplying by s_H^a gives a nonzero section of $E+ar\pi^*P$. Since E is effective and π-exceptional, normality gives $\pi_*\mathcal{O}_S(E)=\mathcal{O}_W$ (rational functions with poles only over codimension at least two are regular on W); the projection formula yields a nonzero section of arP on W, a positive Cartier multiple of P. □

3. Stable torus quotients and invariant sections

Let Z be smooth projective, let $L=-K_Z$ carry a smooth semipositive Hermitian metric, and let the algebraic torus T act on Z. We build a projective quotient with klt stabilizer boundary and a bounded semipositive anti-log-canonical metric. Rational connectedness is used only at the very end, where it gives Euler characteristic one on a resolution of the quotient.

Replace T by its image in $\mathrm{Aut}(Z)$: the kernel acts trivially on T_Z and hence on the natural linearization of L, so invariants are unchanged. For the quotient construction assume $t=\dim T>0$; the trivial action is handled at the end. Average the metric over the compact form $K\simeq(S^1)^t$ of T to obtain a smooth K-invariant semipositive metric h on the same naturally linearized bundle.

3.1. **Weights and an ample perturbation.** For a T-linearized line bundle C, a point $x\in Z$, and an integral one-parameter subgroup $\lambda:\mathbb{C}^*\to T$, set

$$w_C(x,\lambda)=\text{the weight on } C_{x_0},\qquad x_0=\lim_{\tau\to0}\lambda(\tau)x, \tag{14}$$

with the limit existing by projectivity and the weight convention that a fiber vector of weight w transforms by τ^w. For very ample C this is the maximum over eigensections nonvanishing

at x of the pairing with λ. *Sign check:* if s is an eigensection of weight α and l a nonzero lift at x, the scalar of s relative to $\lambda(\tau)l$ is $\tau^{-\alpha}$ times its scalar at x; the smallest order over a generating set is $-\max\alpha$, and a basepoint-free set has minimal order zero, so the transformed lift has order $\max\alpha$, matching (14). In this convention semistability is $w_C(x,\lambda)\geq 0$ for all nonzero one-parameter subgroups, and stability is strict positivity; see [Mumford–Fogarty–Kirwan(1994)] and [Dolgachev–Hu(1998), §1.1.5].

Torus linearization [Sumihiro(1974)], in the forms [Knop–Kraft–Luna–Vust(1989), Proposition 2.4 and Corollary 2.6], gives an equivariantly very ample J; pass to a high power so $J+L$ is also very ample with the natural linearization on L. For each bundle the nonvanishing eigensections at a point form a subset of a fixed finite list, so the weight functions, and the difference

$$w_L = w_{J+L} - w_J,$$

take only finitely many patterns; each is homogeneous and piecewise linear on a finite rational polyhedral fan in $\mathrm{Lie}(T)_{\mathbb{R}}$ (the real cocharacter space).

Lemma 3.1. *On a dense Zariski open subset of Z, the generic weight pattern of L is strictly positive on every nonzero real cocharacter.*

Proof. Fix a nonzero integral cocharacter λ. The Białynicki-Birula decomposition [Białynicki-Birula(1973)] has an open attracting stratum (smooth attracting pieces with the tangent-weight description follow from [Jelisiejew–Sienkiewicz(2019), Theorem 1.5 and Corollary 7.3]). At the limiting fixed component all tangent weights are ≥ 0, and some are positive: otherwise the attracting stratum has no positive-dimensional transverse directions and openness would make λ act trivially, contradicting effectiveness. The weight of $L=\det T_Z$ is the sum of those tangent weights, hence strictly positive on this open stratum. The generic patterns of J and $J+L$ hold on one dense open where every chosen eigensection is nonvanishing; it meets the attracting stratum for each λ, so its weight function is positive for every nonzero integral cocharacter. Since it is a finite union of rational linear pieces, positivity on every nonzero integral cocharacter upgrades to every nonzero real cocharacter (a nonpositive locus on a rational cone would contain a nonzero rational point), and on the unit sphere this continuous homogeneous function has a positive minimum. □

Proposition 3.2. *There are a positive rational δ and a rational character shift of arbitrarily small size for the ample rational linearized bundle $L+\delta J$ such that its stable locus U is nonempty and equals its semistable locus; moreover*

$$w_L(x,\lambda)\geq 0 \quad \textit{for all integral one-parameter subgroups } \lambda,\ \textit{all } x\in U. \tag{15}$$

The shift only defines U; the linearization on L remains natural.

Proof. L is nef, so $L+\delta J$ is ample for every $\delta>0$. Subdivide the finitely many weight fans into one finite pointed rational fan on which every pattern of J and L is linear. For each pattern violating (15), choose an integral ray of negative weight; there are finitely many. For small $\delta>0$ and a small rational character shift all these chosen weights stay negative, so such a pattern cannot occur at a semistable point of the perturbed linearization. Meanwhile Lemma 3.1 and its uniform positive lower bound on the unit sphere keep the generic pattern strictly positive, so the stable locus is nonempty.

Having fixed such a rational δ, choose the character off the finitely many hyperplanes where some pattern vanishes on a ray of the common fan. Then every semistable point has strictly positive weights on those rays, and linearity on pointed cones makes the weights strictly positive on every nonzero cocharacter, so the point is stable. Clearing denominators gives an ordinary ample linearized bundle with the same loci. □

By projective GIT the quotient

$$q : U \longrightarrow W = U/T \tag{16}$$

is a geometric quotient, W is projective, and the action has finite stabilizers on U. Stable equals semistable, so this is the full projective GIT quotient, not an open piece of one.

3.2. **Stabilizer boundary and the anticanonical identity.**

Proposition 3.3. *W is normal and $\mathbb{Q}$-factorial, and there is an effective rational divisor*

$$D = \sum_B \Big(1 - \frac{1}{e_B}\Big) B$$

on W, with e_B the generic ramification index of a finite stabilizer slice over the prime B, such that (W, D) is klt. With $P = -(K_W + D)$,

$$L|_U = q^* P \tag{17}$$

as equivariant $\mathbb{Q}$-line bundles, using the natural linearization on L and the trivial action on the base.

Proof. Use Luna's slice theorem [Luna(1973)] in an invariant affine chart from the ample GIT linearization. Near an orbit with finite stabilizer G there is a smooth slice S_{sl} and, after an étale base localization, a local model $T \times^G S_{\mathrm{sl}} \to S_{\mathrm{sl}}/G$. The slice is smooth as U is smooth with constant orbit dimension. Normality follows from taking invariants of a normal ring. Locally $\mathbb{Q}$-factoriality: a Weil divisor pulls back to a Cartier divisor on the smooth finite cover, and the product of its translates gives a local equation for a positive multiple downstairs; this descends through the étale localizations. Hence W is normal and $\mathbb{Q}$-factorial.

Discarding an ineffective kernel, at a general point over a prime B the finite map has ramification order e_B and ramification divisor coefficient $e_B - 1$, so in codimension one $K_{S_{\mathrm{sl}}} = p^*(K_W + D)$. The e_B are intrinsic (multiplicities of the quotient pullback of B), so the charts glue to one global effective rational divisor D, with only finitely many ramified primes.

Klt: extend any divisorial valuation over W to the function field of the slice. If it has ramification index e, its log discrepancy upstairs (for a smooth variety with zero boundary) is e times its log discrepancy for (W, D) downstairs; the upstairs discrepancies are positive, so the downstairs ones are too. This is the finite-morphism criterion [Kollár–Mori(1998), Proposition 5.20]; the finite-group computation with its ramification boundary is in [Braun–Greb–Langlois–Moraga(2024), Proposition 3.5 and equation (3.4)].

Linearization in (17): on U the infinitesimal action is an injective subbundle $\mathrm{Lie}(T) \otimes \mathcal{O}_U \to T_U$ of rank $\dim T$ (finite stabilizers). Over a dense smooth $W^\circ \subset W$ with $U^\circ = q^{-1}(W^\circ)$ where q is a submersion, the tangent sequence

$$0 \to \mathrm{Lie}(T) \otimes \mathcal{O}_{U^\circ} \to T_{U^\circ} \to q^* T_{W^\circ} \to 0$$

together with the triviality of the adjoint action (a torus) makes a nonzero element of $\det \mathrm{Lie}(T)$ an equivariant trivialization of the orbit determinant; taking determinants identifies $L|_{U^\circ}$ with $q^*(-K_{W^\circ})$ equivariantly. The orders of this rational identification are exactly the ramification orders above, introducing D and no character twist. A prime in U cannot map into codimension ≥ 2 in W (dimension count), so these slice computations account for every prime upstairs. Finitely many slice charts have stabilizer groups of bounded exponent; a common multiple clears denominators and Cartier indices, giving an actual line-bundle isomorphism with trivial fiberwise action. This is an equivariant rational-line-bundle identity, not just a numerical one. □

3.3. **A uniform bound on translated lifts.**

Lemma 3.4 (Orbit bound)**.** *Let $x_0 \in U$ and $l(x)$ a nonvanishing algebraic local frame of L near x_0. Then $\sup_{t \in T} |t \cdot l(x)|_h$ is bounded above on a neighborhood of x_0. The same holds for holomorphic frames and positive tensor powers of L.*

Proof. Step 1 (no poles along arcs). Let $x(s) \to x_0 \in U$, let $l(s)$ be a nonzero holomorphic lift with nonzero value at 0, and let $t(s) \in T$ be meromorphic at 0. Writing $t(s) = \lambda(s)u(s)$ with an integral cocharacter λ and a holomorphic u invertible at 0, replace (x, l) by (ux, ul); the new initial point $u(0)x_0$ still lies in U, so (15) remains available, and it suffices to treat $t(s) = \lambda(s)$.

For rational $r \in [0, 1]$ let $b(r)$ be the order of $\lambda(s^r)l(s)$ against a nonvanishing local frame at the limiting base point (positive means a zero, negative a pole), defined via a common ramified parameter and divided by its degree. By projectivity the limiting base point exists and algebraicity makes the order finite and rational; smoothness and positivity of the metric give

$$b(r) = \lim_{s \downarrow 0} \frac{-\log |\lambda(s^r)l(s)|_h^2}{-\log |s|^2}. \tag{18}$$

For fixed real $s > 0$ the numerator is convex in r: the pullback weight along the holomorphic orbit map $\mathbb{C}^* \to Z$ is psh and, being invariant under the compact circle, convex in the real log coordinate; composing with $r \mapsto r \log s$ and dividing by the positive denominator preserves convexity. Passing to limits makes b convex on rational points, with $b(0) = 0$.

Step 2 (initial slope). We claim $b(r) = r\, w_L(x_0, \lambda)$ for small positive rational r. Use the very ample J and $J + L$ of §3.1: pick a nonvanishing lift to J along the arc and tensor with $l(s)$ to get a lift to $J + L$. For a very ample C with eigensection weights α_j and orders $\nu_j \geq 0$ of the C-lift ($\nu_j = \infty$ if a section vanishes identically), the transformed lift has orders $\nu_j - r\alpha_j$ and, since the sections generate C, order $-\min_j(\nu_j - r\alpha_j)$. For small enough $r > 0$ every minimizer has $\nu_j = 0$, so the order is $r \max_{\nu_j=0} \alpha_j = r\, w_C(x_0, \lambda)$. Subtracting the J-formula from the $J + L$-formula gives the claim.

Step 3 (no pole). Convexity with $b(0) = 0$ gives $b(r) \leq r\, b(1)$, hence $b(1) \geq b(r)/r = w_L(x_0, \lambda) \geq 0$ by (15). So a transformed lift has no pole along any such arc.

Step 4 (uniformity). Suppose $\sup_{t \in T} |t \cdot l(x)|_h$ were unbounded as $x \to x_0$. Compactify T projectively and compactify the total space of L as $\mathbb{P}(\mathcal{O}_Z \oplus L)$ with its infinity divisor. Take the closure of the graph of $(t, x) \mapsto t \cdot l(x)$, keeping x a parameter, in the resulting proper family over the parameter neighborhood. Since Z is projective and h smooth, an unbounded sequence of norms gives a point at fiber infinity above x_0; algebraic curve selection through it, with generic point in the graph, yields after normalization an arc $x(s) \to x_0$ and meromorphic $t(s)$ with $t(s)l(x(s))$ having a pole, contradicting Step 3. Hence the bound is locally uniform.

A holomorphic frame differs from an algebraic one by a nowhere-vanishing holomorphic function, bounded above and below on a smaller neighborhood, and positive powers preserve the estimate. □

3.4. **The orbit-supremum metric.** Fix a divisible positive integer N for which (17) is an isomorphism of line bundles. For $y \in W$ and $v \in (NP)_y$, pull v back to the orbit and set

$$|v|_{h_{NP}} = \sup_{x \in q^{-1}(y)} |q^* v(x)|_{h^N}. \tag{19}$$

The definition is independent of the orbit representative (the isomorphism is equivariant) and scales by $|c|$ in v; the choice of N kills finite-stabilizer characters.

Proposition 3.5. (19) *defines a metric whose Nth root on P has locally bounded psh weights on W, and whose pullback to every smooth resolution also has locally bounded psh weights. In particular P is nef.*

Proof. Boundedness. In a finite slice chart, nearby quotient points have representatives on the slice near a fixed one; Lemma 3.4 bounds the pulled-back frame of NP above, and the value at the representative itself is bounded below in a small chart (smooth metric, nonvanishing frame). So the supremum is finite and bounded above and below in frames, and its negative logarithm is locally bounded.

Plurisubharmonicity. Over a small smooth slice ball B and the universal cover $\mathbb{C}^t \to T$, let e be the pulled-back frame of NP and

$$\Phi(z,\zeta) = -\log|\exp(\zeta)\cdot e(z)|^2_{h^N}, \qquad z \in B,\ \zeta \in \mathbb{C}^t.$$

The action map is holomorphic and the translated frame is holomorphic and nonvanishing, so Φ is jointly psh. It is independent of $\operatorname{Im}\zeta$ by compact-form invariance. The domain $B \times \mathbb{C}^t$ is pseudoconvex with connected tube fibers, so Kiselman's minimum principle [Kiselman(1978)], in the form [Deng–Wang–Zhang–Zhou(2019), Theorem 1.1], makes $\inf_\zeta \Phi(z,\zeta)$ psh or identically $-\infty$; the bounds exclude the latter. This equals the weight of (19). Kiselman gives positivity *after* boundedness; it does not supply the bound.

The slice weights are invariant under the finite slice group with the right pullback transformation, so they descend to locally bounded upper-semicontinuous functions. For positivity, take a holomorphic disc in the quotient, normalize a component of its finite pullback dominating the disc, lift it after a finite ramified disc cover, and use that the pulled-back weight is subharmonic there and hence subharmonic downstairs off the branch points; boundedness extends it across them. The same argument after composing with a disc in a smooth resolution gives subharmonic restrictions, so the pulled-back weights on the resolution are psh (equivalently extend from the regular locus with local upper bounds, [Demailly(2012), Chapter I, Theorem 5.24], on the resolution). Taking the Nth root gives a metric on P.

Nefness. Let C be an irreducible curve on a resolution. A locally bounded psh weight restricts to a psh function on the normalization of C (not identically $-\infty$), and the resulting positive curvature measure has total mass equal to $\deg(P|_C) \geq 0$. So P has nonnegative intersection with every curve on the resolution, and the projection formula gives nefness on W. □

3.5. **The Euler premise and extension.**

Proof of Theorem 1.2. Assume Z is rationally connected. If the image of T is trivial, every section is naturally invariant and Theorem 1.1 applied to $(Z,0)$ gives a nonzero one, since L is nef with a smooth semipositive metric and $\chi(Z,\mathcal{O}_Z) = 1$.

Otherwise take the quotient (W,D) of Proposition 3.3, $\mathbb{Q}$-factorial and klt with effective rational boundary; Proposition 3.5 gives nefness and the bounded metric for $P = -(K_W + D)$.

Let $\widetilde{W} \to W$ be a smooth projective resolution. The rational map $Z \dashrightarrow \widetilde{W}$ is dominant; resolve its indeterminacy to a morphism $\widetilde{Z} \to \widetilde{W}$. Rational connectedness is preserved by smooth projective birational modification and by dominant rational maps to smooth projective targets [Kollár(1996)]; equivalently, pullback of forms under the dominant map is injective with $H^0(\widetilde{Z},\Omega^p) = 0$ for $p > 0$, so $H^0(\widetilde{W},\Omega^p) = 0$ for $p > 0$ and Hodge symmetry yields $\chi(\widetilde{W},\mathcal{O}_{\widetilde{W}}) = 1$ (if W is a point, directly).

Theorem 1.1 gives a nonzero section σ of a positive Cartier multiple mP; take a power so m is also divisible by the integer of (17). Its pullback $\widetilde{\sigma}$ is a nonzero invariant section of mL on U. The bounded quotient metric makes $|\sigma|$ bounded on the compact W; (19) then gives

$$|\widetilde{\sigma}(x)|_{h^m} \leq |\sigma(q(x))|_{h_{mP}} \leq C \qquad (x \in U). \tag{20}$$

For $x \in Z \setminus U$, pick a holomorphic nonvanishing frame of mL on a small coordinate neighborhood; smoothness of h bounds the frame's norm above and below, so the coefficient of $\widetilde{\sigma}$ is a bounded holomorphic function off the algebraic set $Z \setminus U$. The removable-singularity theorem extends it across $Z \setminus U$ [Demailly(2012), Chapter I, Corollary 5.25], in particular at the generic point of every omitted divisor; normality handles the codimension-two remainder. Unique local extensions glue to a global section of mL.

For $t \in T$, the translated global section agrees with the original on the dense open U, hence everywhere, and it is nonzero because the original is nonzero on U. This is the invariant nonvanishing. □

Remark 3.6. Natural linearization cannot be dropped even on $\mathbb{P}^1$. For $t \cdot [z_0 : z_1] = [tz_0 : z_1]$ the weights on $H^0(\mathbb{P}^1, -mK_{\mathbb{P}^1})$ run from $-m$ to m under the natural linearization. Tensoring the linearization on $-K_{\mathbb{P}^1}$ by the character $t \mapsto t^2$ shifts the weights in the mth power to run from m to $3m$, so every positive power has zero invariant subspace. This is why the auxiliary character used to choose the stable open is not transferred to L.

4. The smooth projective application

A section on a compact factor of the universal cover descends only if it respects the deck action. The invariant statement of Theorem 1.2 supplies exactly that compatibility with the residual torus monodromy, giving another proof of the smooth nonvanishing result of the companion article [OpenAI(2026), Theorem "Anticanonical nonvanishing"], whose finite-volume proof is replaced here by the bounded-metric criterion and stable quotient. From the companion we take only the finite-cover structure and the finite-étale norm.

Corollary 4.1. *Let X be a smooth connected projective complex variety with smoothly semipositive anticanonical bundle. Then $H^0(X, -mK_X) \neq 0$ for some integer $m > 0$.*

Proof. Use the companion's [OpenAI(2026), Theorem "Finite cover with compact torus monodromy"] in its form retaining the deck action: there is a finite étale cover $X' \to X$ whose universal cover is

$$\widetilde{X} = \mathbb{C}^q \times B \times Z,$$

with compact projective factors, K_B trivial, and Z rationally connected with smoothly semipositive anticanonical bundle (either factor may be a point). The deck group acts by translations on $\mathbb{C}^q$, trivially on B, and on Z through a compact torus inside an algebraic torus T. The Euclidean and B factors carry invariant nowhere-zero canonical frames.

By Theorem 1.2 choose a nonzero T-invariant section s_Z of $-mK_Z$ (a constant section if Z is a point). Tensor with the mth powers of the dual canonical frames on the other two factors to get a nonzero section of $-mK_{\widetilde{X}}$ invariant under the deck group of $\widetilde{X} \to X'$; the natural linearization is essential, since the action on the anticanonical product is the differential one used in Theorem 1.2. The section descends to X' and is algebraic by projectivity. If d is the degree of $X' \to X$, the companion's finite étale norm [OpenAI(2026), Lemma "Finite étale norm"] gives a nonzero section of $-mdK_X$. □

The geometric input rests on Yau's prescribed-Ricci theorem [Yau(1978)] and the structure results of Demailly–Peternell–Schneider [Demailly–Peternell–Schneider(1996)] and Campana–Demailly–Peternell [Campana–Demailly–Peternell(2015), Theorem 1.4]. A finite cover need not be a product: the residual torus monodromy is why an invariant section on Z, not an arbitrary one, is needed.

The positive multiple in Corollary 4.1 cannot generally be reduced to the first power. For an Enriques surface S, the bundle K_S is nontrivial of order two and $H^0(S, -K_S) = 0$ [Dolgachev(2016)]. A flat metric on $-K_S$ and the Fubini–Study metric on $\mathcal{O}_{\mathbb{P}^1}(2)$ make the anticanonical bundle of $S \times \mathbb{P}^1$ smoothly semipositive; the product formula for sections shows it has no first-power section while its second power does. Its restriction to a $\mathbb{P}^1$ fiber has degree two, so this example is not explained by torsion of the anticanonical bundle itself.

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